Ankit Kumar

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Summary

Robotics MS student specializing in perception, SLAM, and autonomous navigation for drones and mobile robots. Experienced in stereo vision, LiDAR localization, ROS2, Nav2, embedded deployment, and real-time robotic systems. Skilled in developing mapping, navigation, and autonomy solutions for intelligent robotic platforms.

Education

University of Minnesota — Twin Cities, MN
Masters of Science in Robotics, GPA: 3.78/4.0 — Sept 2024 – May 2026
Coursework: Machine Learning, Intelligent Robotic system, Robot Vision, Deep Learning

Jadavpur University — Kolkata, India
Bachelors of Mechanical Engineering, GPA- 8.55/10 — Aug 2018 – May 2022

Technical Skills

Programming: Python, C++, MATLAB
Robotics & Systems: ROS/ROS2, Nav2, MAVROS, ArduPilot, Gazebo (Classic/Harmonic), NVIDIA Isaac Sim
Perception SLAM: Stereo Vision, Visual SLAM, RTAB-Map, Camera Calibration, LiDAR Localization, Optical Flow, Triangulation
Machine Learning: PyTorch, YOLO, CNNs, Object Detection & Tracking, 3D Reconstruction
Embedded Systems: Jetson Orin Nano, TensorRT, Linux, GStreamer, Git

Experience

Flow Field Imaging Lab (FFIL), University of Minnesota — Twin Cities, MN
Research Assistant — May 2025 – Present

- Developed an end-to-end real-time perception and control pipeline on a custom drone platform using synchronized stereo cameras and onboard Jetson Orin Nano for wildfire ember tracking.
- Designed and trained a multi-head U-Net with custom multi-task loss, achieving 90–95% detection accuracy and improving small-object recall under motion blur and low-contrast conditions.
- Developed a calibrated stereo vision pipeline with triangulation and temporal association, achieving <10 cm 3D trajectory error for real-time ember tracking and motion estimation.
- Integrated perception with control loop and optimized for embedded deployment, achieving 10 FPS on Jetson Orin Nano.
- Evaluated system under varying conditions, improving detection robustness by 20–30% using synthetic data augmentation.

Projects

Autonomous Mobile Robot Navigation SLAM in Isaac Sim| ROS2, RTAB-Map, Nav2, LiDAR — Jan 2026 - May 2026

- Developed a multi-sensor autonomous mobile robot pipeline in NVIDIA Isaac Sim integrating RTAB-Map RGB-D SLAM, LiDAR localization, and Nav2 waypoint navigation using ROS2.
- Implemented a custom 4-wheel differential drive controller and improved SLAM localization consistency, achieving ∼0.82 m ATE RMSE and ∼0.19 m RPE RMSE using evo trajectory evaluation.
- Integrated occupancy-grid mapping, AMCL localization, and Dijkstra-based path planning for autonomous indoor navigation.

3D Reconstruction Using Autonomous Drone Imagery| ROS, OpenCV, Pytorch, Instant NGP — Sep 2024 - Dec 2024

- Developed an autonomous drone algorithm that detected and circled target objects along optimized viewpoints for high-resolution multi-view image capture.
- Built a 3D reconstruction pipeline using COLMAP and Instant-NGP, implementing image-quality filtering, target cropping, and optimized hash-grid encoding for high-fidelity scene reconstruction.

GZ-Sim with ArduPilot for VTOL Simulation| Gazebo Harmonic, ROS 2, ArduPilot — May 2025 - Aug 2025

- Developed a Gazebo Harmonic–ArduPilot SITL simulation framework for VTOL autonomous flight testing using ROS2, MAVROS, and MAVLink communication.
- Executed autonomous waypoint missions through QGroundControl, enabling fixed-wing navigation, trajectory following, and closed-loop flight validation in simulated environments.

AI-Powered Football Player Tracking & Analysis| Pytorch, OpenCV, YOLO, KMeans Clustering — Oct 2024 - Dec 2024

- Developed a YOLOv8-based detection and tracking system for dynamic scenes, detecting multiple classes in video streams.
- Used KMeans clustering for team classification and optical flow for camera movement analysis.
- Applied perspective transformation to convert image-space motion into real-world distance and velocity estimates.